

Comparative and Explainable Network Intrusion Detection on CICIDS2017

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Abstract

This report presents a comparative study of machine learning classifiers for network intrusion detection using the CICIDS2017 dataset. Rather than training a single model and reporting its accuracy, the study builds a controlled experimental pipeline in which seven classifiers are trained and evaluated under identical preprocessing, splitting, and scoring conditions across a seven-class traffic classification task (Normal Traffic, DoS, DDoS, Port Scanning, Brute Force, Web Attacks, Bots). The study is extended in two directions beyond a standard model comparison. Firstly, four complementary feature-selection techniques i.e. Variance Threshold, Mutual Information, ANOVA F-test, and Recursive Feature Elimination are combined into a consensus feature set, and the effect of dimensionality reduction on each classifier is measured directly. Secondly, the strongest model is interpreted using SHAP to identify which network-flow characteristics drive its predictions, both globally and for individual correctly and incorrectly classified flows. To guard against drawing conclusions from a single lucky train/test split, the top three models are re-evaluated across five random seeds, and the resulting differences are tested for statistical significance using the Wilcoxon signed-rank test with Holm correction. The main findings are that tree-based ensembles substantially outperform linear and probabilistic baselines on this dataset, that a carefully chosen 22-feature consensus subset preserves ensemble performance almost exactly while cutting dimensionality by more than half, that this benefit does not extend to Logistic Regression or Naive Bayes, and that the observed ranking among the top three ensemble models is not statistically significant across repeated runs despite visible differences in mean scores.

1. Objectives

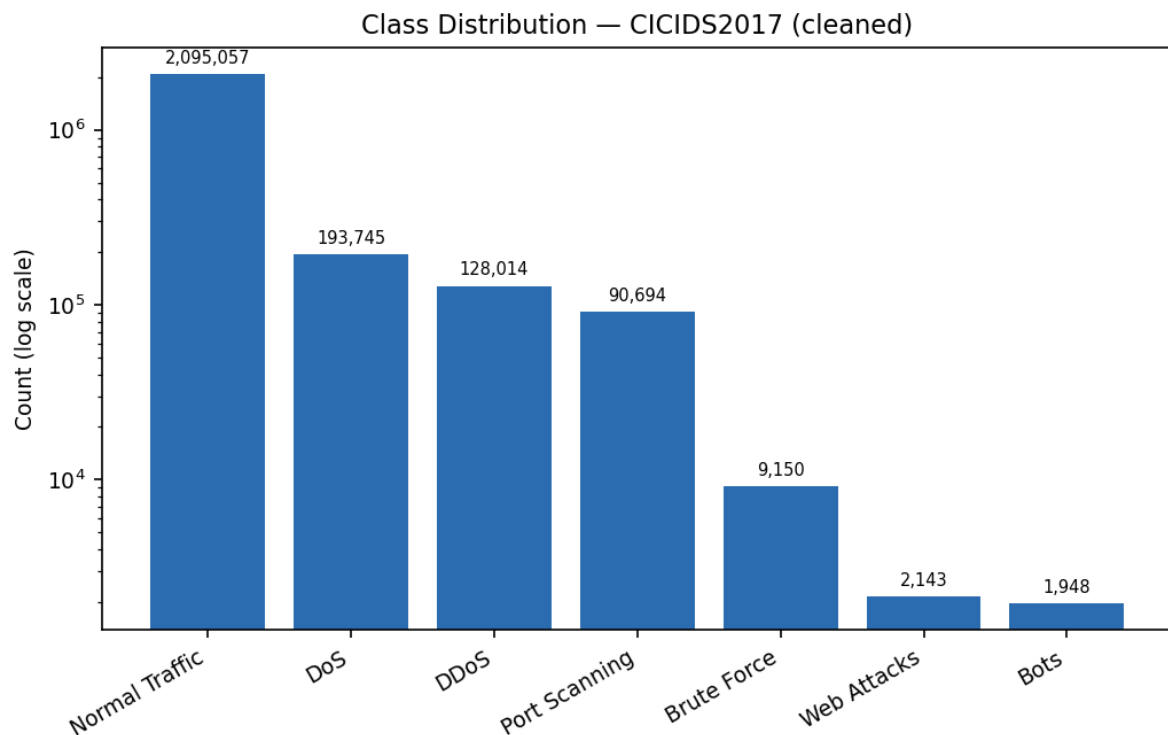
- Compare a broad set of classical machine-learning classifiers for multiclass network intrusion detection under identical experimental conditions.
- Determine whether feature-selection methods can reduce the dimensionality of network-flow data without sacrificing detection performance, and whether the choice of method matters as much as the number of features retained.
- Identify which network-flow features most influence the predictions of the strongest model, using SHAP, and relate these to the features favoured by the feature-selection stage.
- Establish whether performance differences between the top-performing models are stable across repeated experiments and statistically meaningful, rather than an artifact of one particular data split.

- Report computational cost (training and inference time) alongside predictive performance, since a model that is only marginally more accurate but far slower may not be the more practical choice for intrusion detection.

2. Dataset

The study uses a pre-processed CICIDS2017 CSV file, containing 2,520,751 labelled network flow records across 53 columns (52 numeric flow-level features extracted using CICFlowMeter, plus a categorical label). The label distribution is heavily imbalanced:

Attack Type	Count	Share
Normal Traffic	2,095,057	83.11%
DoS	193,745	7.69%
DDoS	128,014	5.08%
Port Scanning	90,694	3.60%
Brute Force	9,150	0.36%
Web Attacks	2,143	0.09%
Bots	1,948	0.08%



The dataset was inspected for missing values, infinite values, and duplicates prior to modelling. No missing values were present. Infinite values were found in rate-based columns (e.g. Flow Bytes/s), and a small number of rows contained physically impossible

negative values in duration, inter-arrival-time, and header-length columns. After replacing infinities with NaN, removing rows with impossible negative values, and dropping 161 exact duplicate rows, the cleaned dataset contained 2,517,681 rows. A reduction of 3,070 rows, or 0.122% of the original data. This is a minor cleaning step, consistent with the dataset's known but limited data-quality issues.

3. Methodology

3.1. Preprocessing and data splitting

The cleaned dataset was split into training (80%) and test (20%) subsets using stratified sampling to preserve class proportions, with a fixed random seed for the main experiment. Feature scaling (`StandardScaler`) was fit exclusively on the training data and then applied to the test data, and all feature-selection procedures described below were likewise fit only on the training partition. This ordering was enforced throughout to avoid leaking test-set information into any preprocessing or selection step.

3.2. Classifiers

Seven classifiers were evaluated under identical conditions: ***Logistic Regression***, ***Decision Tree***, ***Random Forest***, ***Extra Trees***, ***Gradient Boosting***, ***K-Nearest Neighbours***, and ***Gaussian Naive Bayes***. Default or lightly tuned hyperparameters were used across all models to keep the comparison fair, in line with the study's aim of comparing algorithmic families rather than conducting an exhaustive tuning exercise.

3.3. Feature selection

Four complementary feature-selection techniques were applied to the training data:

- **Variance Threshold** (threshold = 0.0), to identify constant or near-constant features.
- **Mutual Information**, to rank features by their statistical dependency with the target label.
- **ANOVA F-test** (`SelectKBest`), evaluated additionally across $k = 10, 20, 30$, and 50 to observe how performance scales with the number of retained features.
- **Recursive Feature Elimination (RFE)**, using a Decision Tree estimator, reduced to 20 features.

The top-20 (or selected) feature sets from Mutual Information, ANOVA, and RFE were then combined into two derived sets: a strict **consensus set** (features appearing in all

three rankings) and a **majority-vote set** (features appearing in at least two of the three), used for the reduced-feature experiments described in Section 4.2.

3.4. Explainability

SHAP (`TreeExplainer`) was applied to the strongest model, Random Forest, retrained on the majority-vote feature set. To keep computation tractable, SHAP values were computed on a stratified sample of the test set (approximately 3,474 flows, with a floor of 150 samples per class to ensure the rarest categories were represented). Global feature importance, a multiclass bar summary, a class-specific beeswarm plot, and four local waterfall explanations were produced, covering a correctly classified benign flow, a correctly classified attack flow, a false negative, and a difficult attack-versus-attack misclassification.

3.5. Repeated experiments and statistical testing

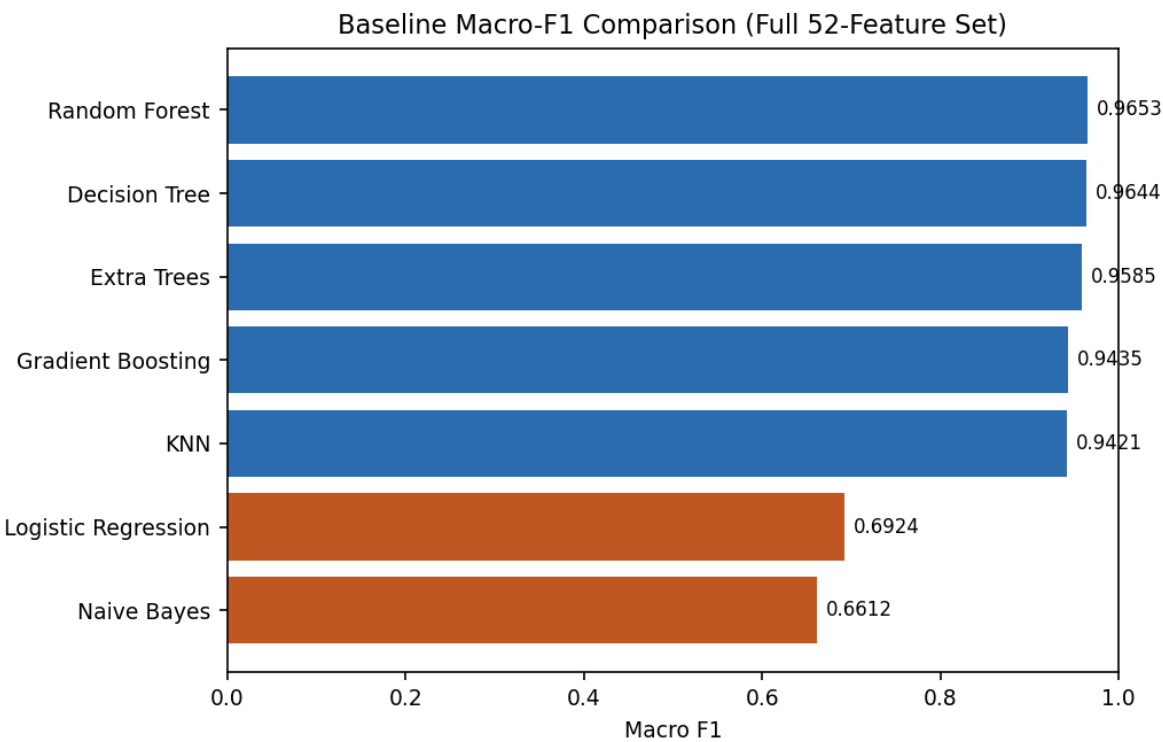
To assess the stability of the model ranking, the top three models by macro-F1 (Random Forest, Extra Trees, Decision Tree) were retrained on the majority-vote feature set across five independent stratified splits (seeds 42, 52, 62, 72, 82), each with its own scaler fit on that split's training data. The resulting macro-F1 scores were compared pairwise using the Wilcoxon signed-rank test, with Holm correction applied across the three comparisons to control the family-wise error rate.

4. Results

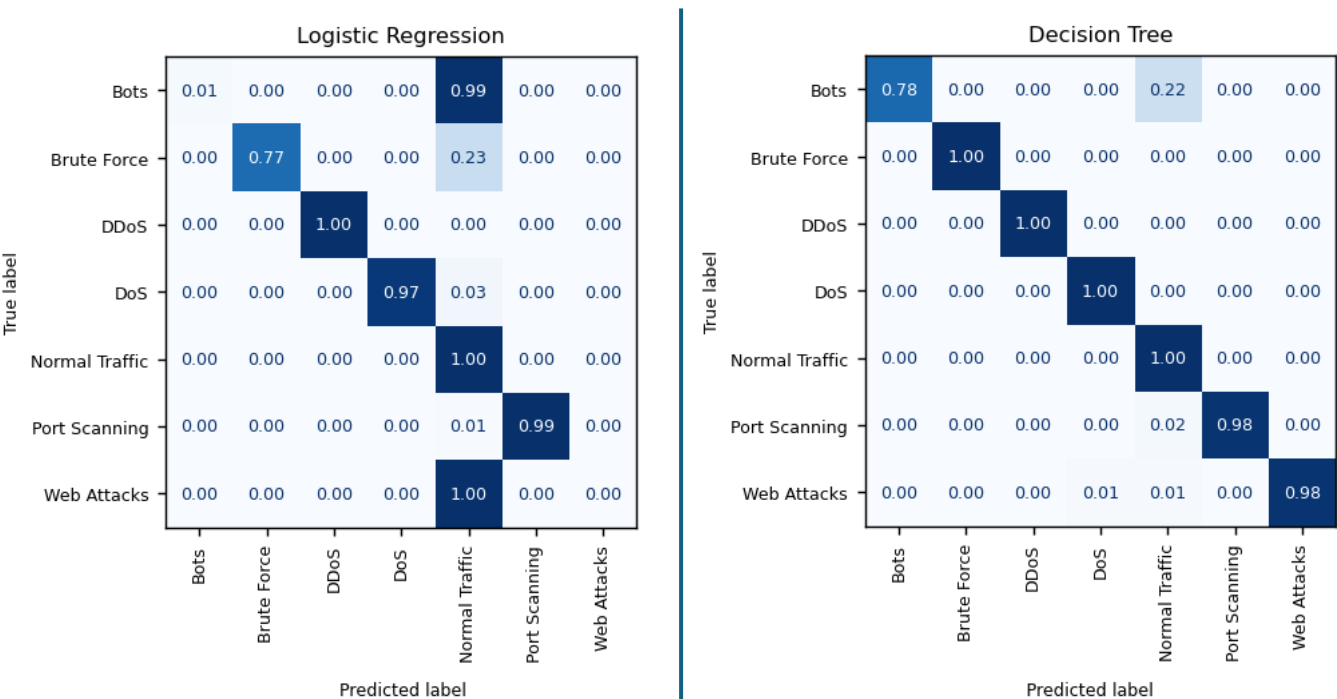
4.1. Baseline comparison (full 52-feature set)

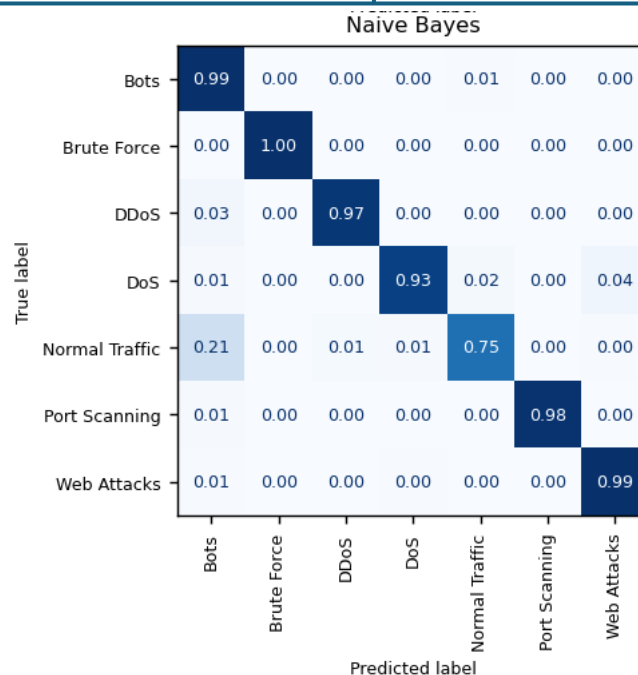
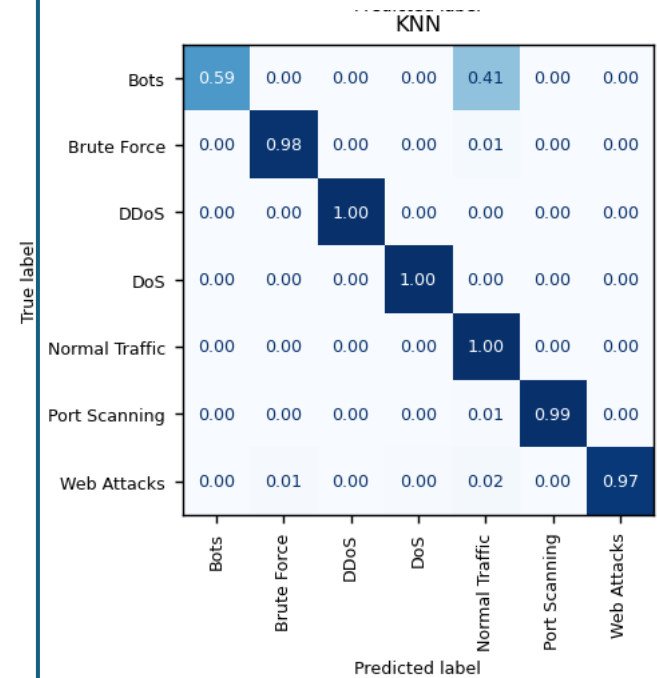
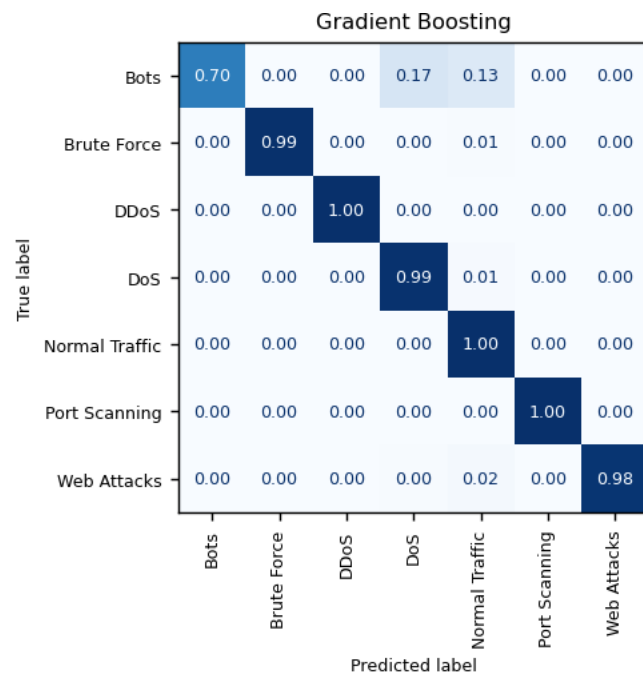
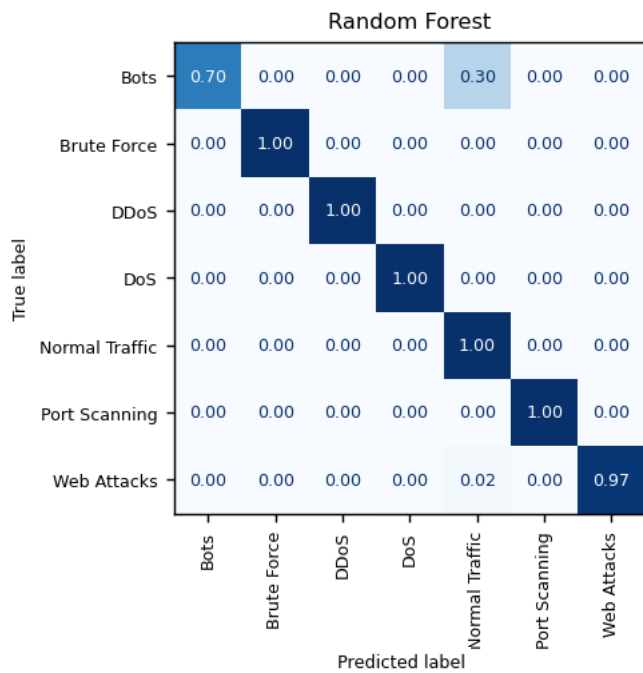
Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	F1 (weighted)	Inference Time (s)
Random Forest	0.9989	0.9811	0.9526	0.9653	0.9989	2.78
Decision Tree	0.9983	0.9662	0.9626	0.9644	0.9983	0.16
Extra Trees	0.9981	0.9711	0.9481	0.9585	0.9981	2.88
Gradient Boosting	0.9978	0.9369	0.9510	0.9435	0.9978	1.34
KNN	0.9979	0.9537	0.9328	0.9421	0.9978	3025.76
Logistic Regression	0.9930	0.7628	0.6760	0.6924	0.9921	0.17
Naive Bayes	0.7878	0.6345	0.9446	0.6612	0.8666	2.15

The gap between accuracy and macro-F1 for Logistic Regression and Naive Bayes is the clearest illustration of why accuracy alone is unsuitable on this dataset, both models exceed 78% accuracy while scoring far lower on macro-F1, which weights every class including the three attack categories that together make up under 0.5% of the data equally.



Confusion Matrices by Model (row-normalized, i.e. per-class recall)

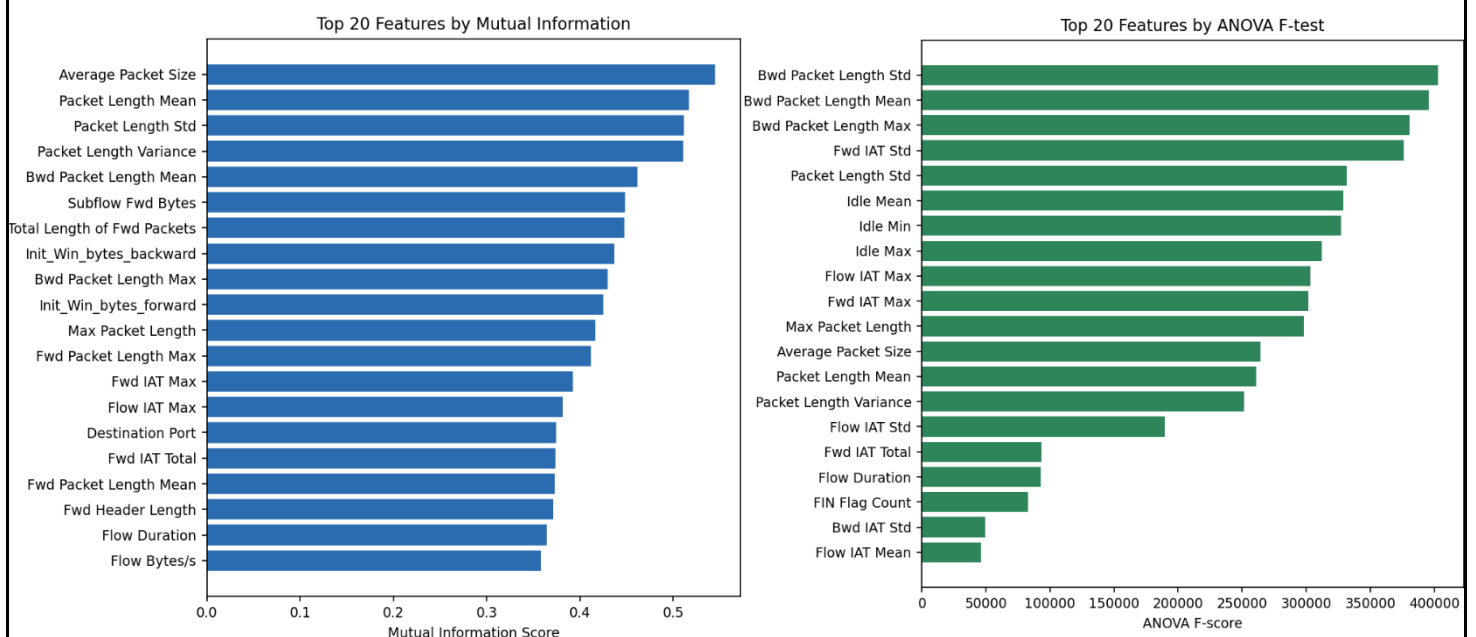


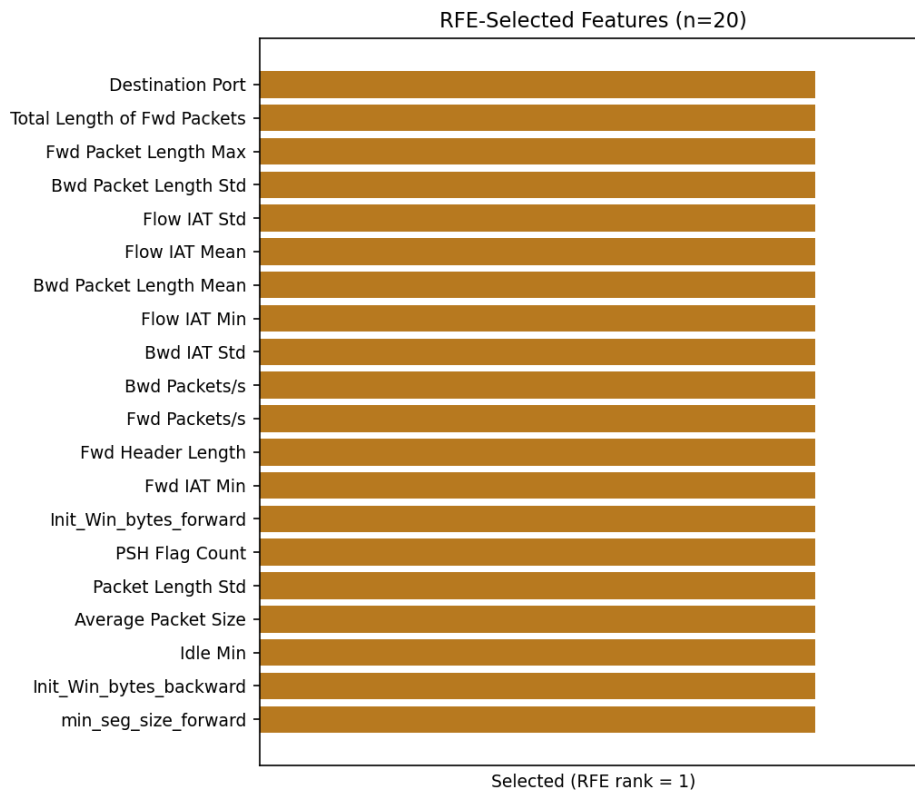


The confusion matrices surface a class-level weakness that **‘Bots’ is the hardest class for every model except Naive Bayes**, with recall ranging from 0.59 (KNN) to 0.78 (Decision Tree), and Random Forest itself only reaching 0.70. In every case, misclassified Bots flows are predicted only as Normal Traffic, Gradient Boosting is a partial exception, splitting its Bots errors between Normal Traffic (17%) and DoS (13%). This is consistent with the intuition that botnet traffic is designed to resemble ordinary background traffic rather than an aggressive attack signature, and it echoes a near-identical finding reported for this same dataset elsewhere (Bots recall of 0.72 for Random Forest). Naive Bayes is the one model that detects Bots well (0.99 recall), but only because its decision boundary is skewed broadly toward attack classes at the expense of Normal Traffic recall (0.75) and overall macro-F1, it is not a genuinely better detector of Bots so much as a less discriminating one overall.

4.2. Feature selection

Variance Threshold removed no features; all 52 retained non-zero variance, confirming that this method functions only as a weak sanity filter on this dataset rather than a meaningful reduction technique. Mutual Information and the ANOVA F-test each produced a ranked list of features, with Average Packet Size, Packet Length Mean/Std/Variance, and Bwd Packet Length Mean consistently appearing near the top of both. RFE, using a Decision Tree estimator, converged on a 20-feature subset dominated by header-length, inter-arrival-time, and window-size features.





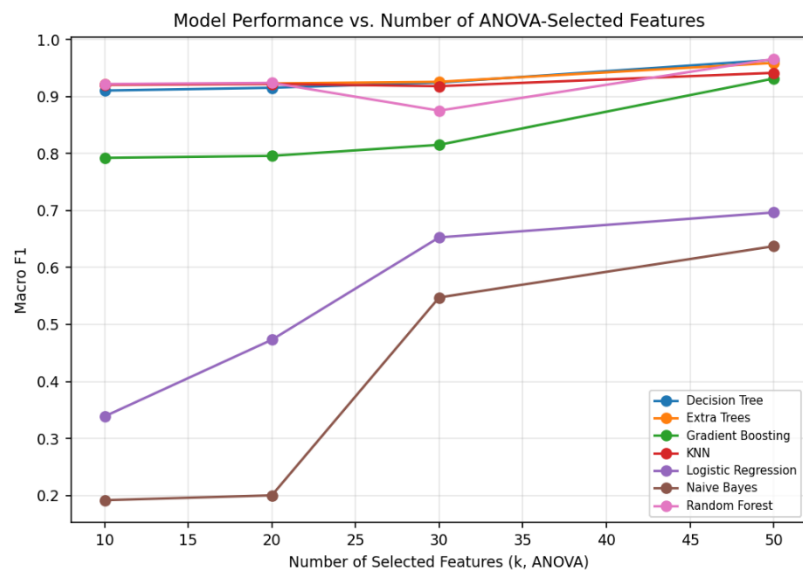
The intersection of all three rankings produced only three consensus features (Average Packet Size, Bwd Packet Length Mean, Packet Length Std), while relaxing the criterion to a majority vote (agreement across at least two of the three methods) produced a 22-feature subset, a reduction of 58% relative to the full feature set. This majority-vote set was carried forward into a second round of model evaluation:

Model	N Features	Accuracy	F1 (macro)	Train Time (s)	Inference Time (s)
Random Forest	22	0.9990	0.9659	281.16	1.83
KNN	22	0.9987	0.9653	—	2389.35
Decision Tree	22	0.9983	0.9637	100.82	0.10
Extra Trees	22	0.9984	0.9635	185.29	4.18
Gradient Boosting	22	0.9967	0.8560	25.62	1.39
Logistic Regression	22	0.9452	0.5883	575.60	0.10
Naive Bayes	22	0.4494	0.4503	6.60	0.98

The most notable result of the feature-selection stage is not the reduction itself but a comparison across methods at similar feature counts. A single-method ANOVA top-20 selection produced a Random Forest macro-F1 of only 0.9238, well below the 0.9659 achieved with the 22-feature majority-vote set. Since both subsets are of comparable size, this gap indicates that the choice of feature-selection strategy matters considerably more than the raw number of features retained.

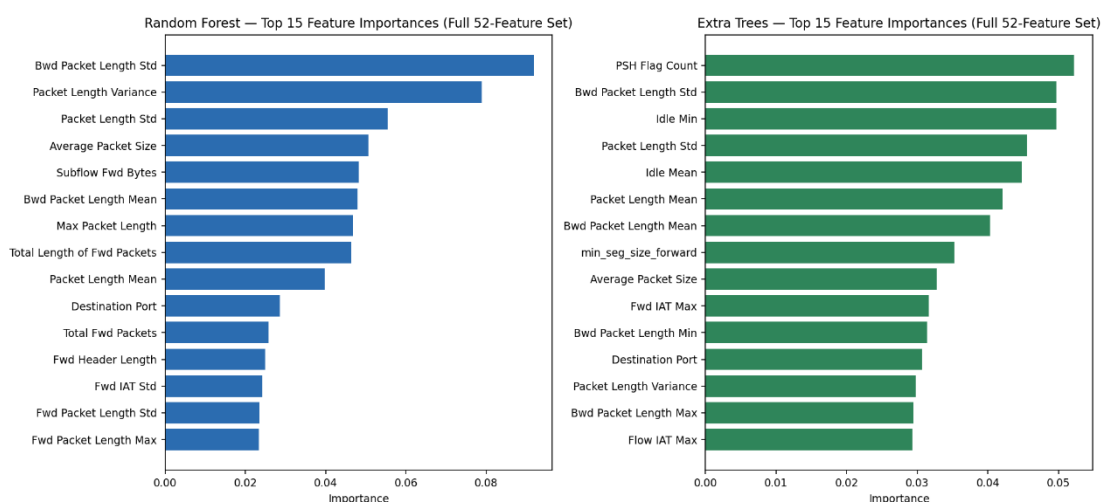
The reduction also affected model families unevenly. Random Forest, Extra Trees, Decision Tree, and KNN retained essentially their full-feature performance on the 22-

feature set. Gradient Boosting, Logistic Regression, and Naive Bayes did not, their macro-F1 scores fell from 0.9435 to 0.8560, from 0.6924 to 0.5883, and from 0.6612 to 0.4503 respectively. A plausible explanation is that the majority-vote set is dominated by closely related packet-length and packet-size



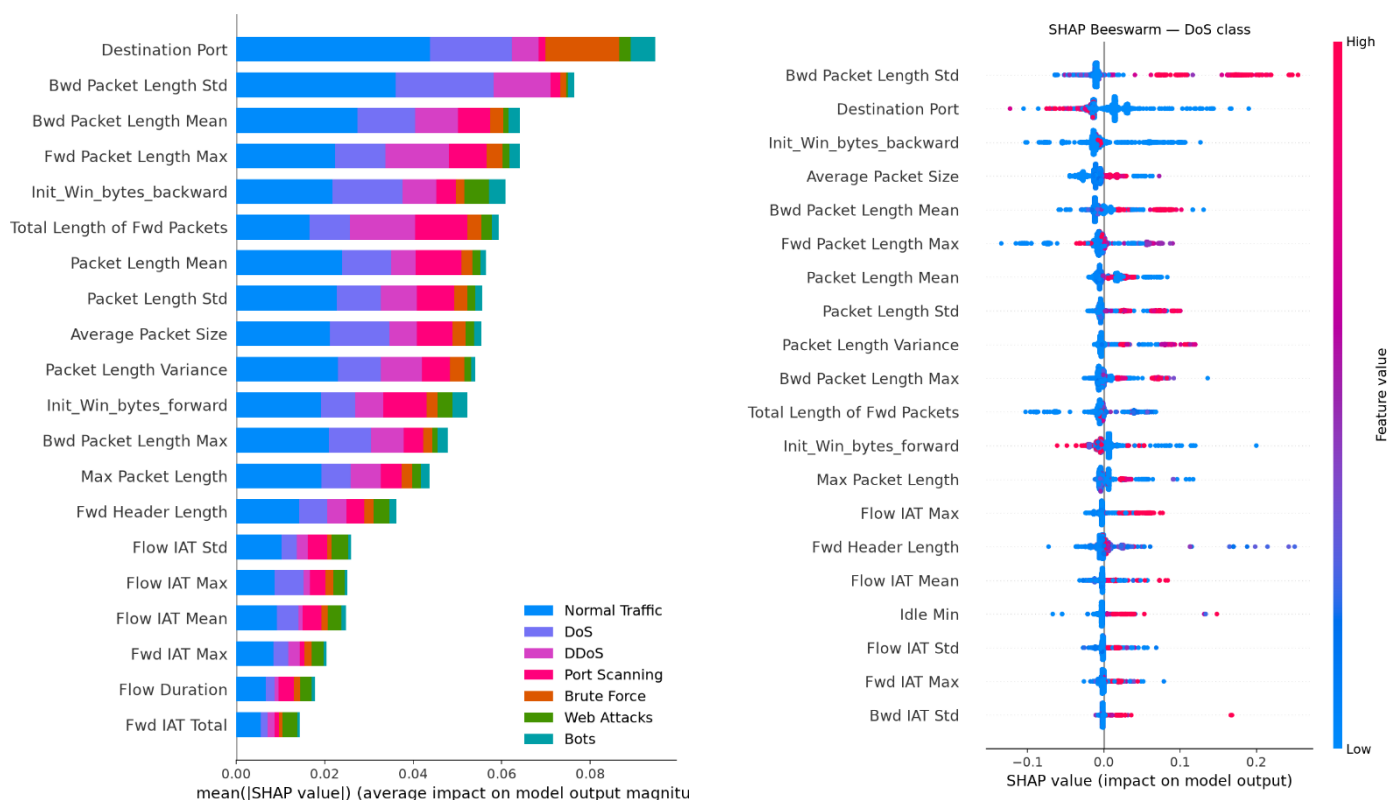
statistics. This redundancy is largely harmless for tree-based splits but works against Naive Bayes' conditional-independence assumption and appears to have made the classification boundary harder for a linear model to separate.

For a full view of what tree ensembles rely on before any feature reduction is applied, Figure given below shows the raw `feature_importances_rankings` for Random Forest and Extra Trees on the complete 52-feature set. Both converge on the same packet-length and packet-size cluster identified by Mutual Information and ANOVA above, which is the first indication that this feature group is not an artifact of any single ranking method.

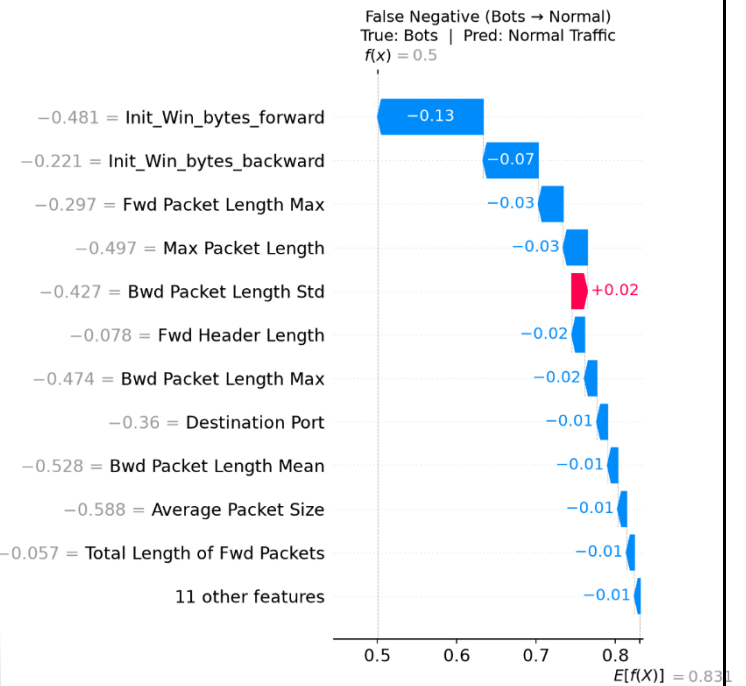
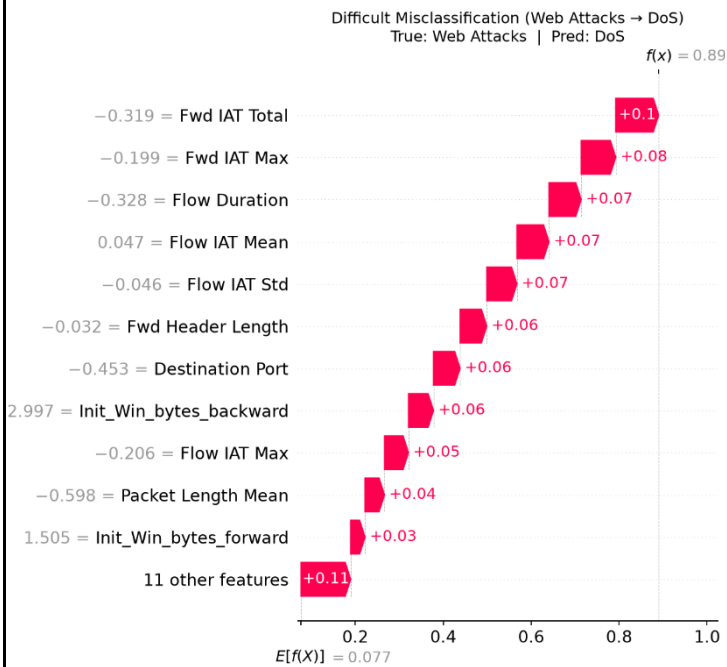
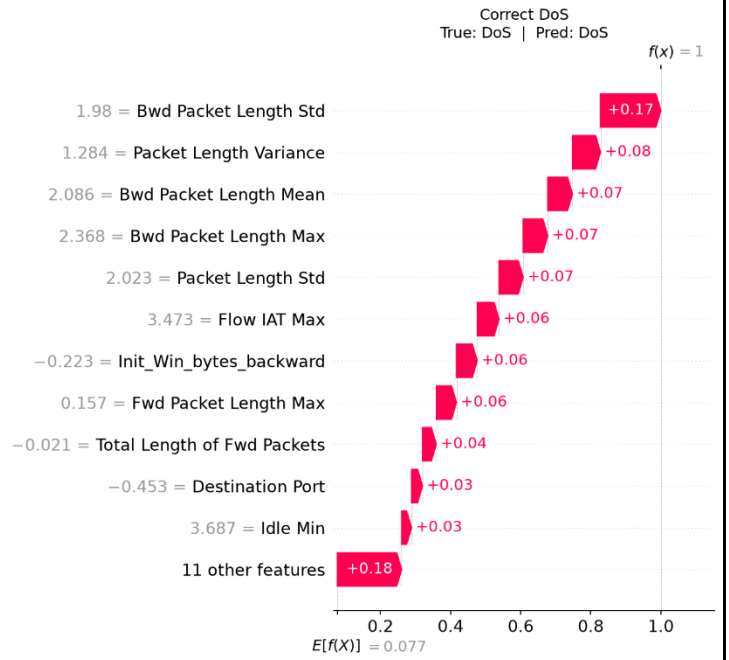
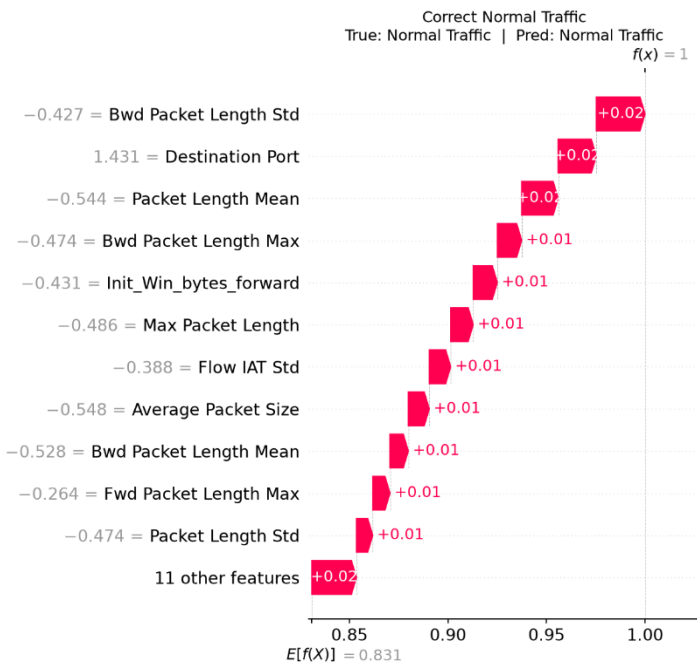


4.3. Explainability

Random Forest, retrained on the 22-feature majority-vote set, reproduced its expected macro-F1 of 0.9659, confirming the retrained model was consistent with the cached result. SHAP's global mean absolute value ranked Destination Port, Bwd Packet Length Std, Bwd Packet Length Mean, Fwd Packet Length Max, and Init_Win_bytes_backward as the five most influential features overall, closely mirroring the packet-length and window-size cluster already surfaced by the feature-selection stage.



Local explanations were generated for four representative cases drawn from the SHAP sample, a correctly classified Normal Traffic flow and a correctly classified DoS flow, both predicted with a full confidence margin, a false negative in which a Bots flow was misclassified as Normal Traffic with a near-zero confidence margin (0.0008), indicating the model was essentially undecided rather than confidently wrong, and a difficult attack-versus-attack case in which a Web Attacks flow was misclassified as DoS with a comparatively high margin (0.82). No false positive existed within the 2,492 Normal Traffic instances contained in the stratified SHAP sample, consistent with the model's very high precision on the majority class observed in the confusion matrix.



4.4. Repeated experiments and statistical significance

Across five random seeds, using the 22-feature majority-vote set, the three strongest models produced the following macro-F1 scores:

Model	Mean F1 (macro)	Std. Dev.	Mean Train Time (s)	Std. Dev.
Random Forest	0.9699	0.0034	324.75	15.93
Extra Trees	0.9670	0.0027	147.46	9.88
Decision Tree	0.9629	0.0022	120.44	20.47

Pairwise Wilcoxon signed-rank tests on the paired per-seed scores gave uncorrected p-values of 0.0625 (Random Forest vs. Extra Trees), 0.0625 (Random Forest vs. Decision Tree), and 0.125 (Extra Trees vs. Decision Tree). After Holm correction, none of the three comparisons reached statistical significance (all adjusted $p \approx 0.19$). Random Forest achieves the highest mean score and the tightest advantage over its nearest

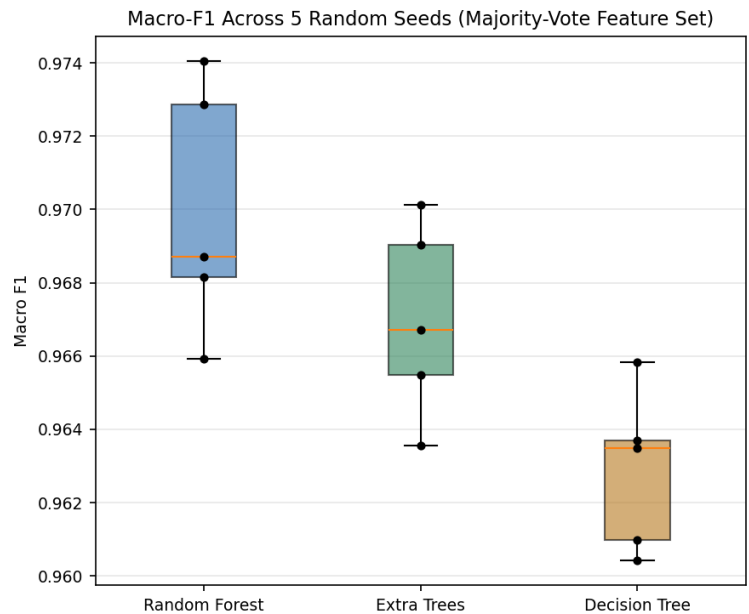
competitor is consistent across all five seeds, but with only five repetitions the test does not have the power to declare this difference statistically reliable.

5. Discussion

The baseline comparison shows a clear pattern also seen in many intrusion-detection studies. Tree-based models are better at handling the complex relationships between network-flow features than linear or simple probabilistic models. This difference becomes clearer when macro-F1 is considered instead of accuracy. Random Forest performed better than Logistic Regression, which suggests that the relationship between the network features and attack types is not simply linear.

The feature-selection results provide one of the most useful findings of the study. The number of features was reduced by 58%, but the best models showed almost no loss in macro-F1. However, using the same number of features selected by only one method gave noticeably worse results. This shows that the way features are selected is more important than simply deciding how many features to keep. The improvement was not the same for all models. Naive Bayes and Logistic Regression performed worse after feature reduction, possibly because the remaining features were mainly related to packet sizes and other correlated values, which are more suitable for tree-based models.

The SHAP analysis supports the feature-selection results. Many of the same packet-length and window-size features appeared as important in both analyses. This increases confidence that these features contain useful information for detecting different attack types rather than being selected by chance. The false-negative example is also important. The misclassified Bots flow had a very small confidence difference between the predicted and actual classes. This means the model was unsure about the decision, rather than being confidently wrong.



The computational results are also important when considering real-world use. KNN achieved a competitive macro-F1 score, but it took around 40 minutes to classify about half a million test flows. This makes it impractical for real-time detection. Random Forest, on the other hand, achieved the highest macro-F1 while taking only a few seconds for inference. Therefore, it provides a better balance between detection performance and computational cost.

Finally, the statistical testing result should also be considered. Although Random Forest had the highest average score, the differences between the top three models were not statistically significant across the five repetitions. This does not mean that the models perform exactly the same. It simply means that, with the current experiment, there is not enough evidence to confidently say that one of them is better than the others.

6. Limitations

- SHAP-based explainability was applied only to Random Forest. Extra Trees and Decision Tree, though close competitors, were not separately interpreted.
- Only the DoS class received a full SHAP beeswarm visualisation; extending this to other minority classes (Port Scanning, Brute Force) would give a more complete class-specific picture.
- As with other CICIDS2017-based studies, this evaluation uses a random stratified split rather than a temporal, session-aware split. Flows generated by the same attack tool during the same session could plausibly appear on both sides of the split, which is a known characteristic of this dataset rather than an error specific to this pipeline.
- Feature importance and SHAP attribution reflect statistical association with the model's predictions, not a causal account of attacker behaviour, and should be read accordingly.

7. Conclusion

Under a controlled, leakage-free experimental pipeline, tree-based ensembles, Random Forest in particular, clearly outperform linear and naive-probabilistic classifiers for multiclass intrusion detection on CICIDS2017, once evaluation moves past raw accuracy to macro-averaged metrics. A consensus-based feature-selection approach, combining Mutual Information, ANOVA, and RFE, reduces the feature space by more than half without a meaningful loss in ensemble performance, and does so considerably more effectively than a single-method reduction of the same size, indicating that the selection strategy, not merely the feature count, is the variable that matters. SHAP analysis on the best-performing model corroborates the feature-selection findings, converging on the same cluster of packet-length and window-size statistics as the dominant predictive signal. Repeated experiments show this ranking is directionally consistent across five random splits, though not yet statistically distinguishable at that sample size, and

computational-cost measurements make clear that raw predictive accuracy alone is an incomplete basis for choosing a model intended for practical deployment.

References

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